

Multi-class Motor Imagery Classification Method Using One-dimensional Convolutional Neural Networks

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Abstract— Motor Imagery-based Brain-computer interface (MI-BCI) enables communications between the human brain and external machines, which is considered a valuable frontier technology. Motor imagery (MI) identification is a fundamental problem for these applications. However, because of the non-stationary and non-linear properties of EEG motor imagery signals, it is very hard to get a high accuracy multi-class method. In this study, a multi-class classification method of MI EEG signals using wavelet packets and a one-dimensional convolutional neural network is presented. Data normalization and wavelet threshold denoising were performed for signal preprocessing. Then EEG signals were decomposed and reconstructed by wavelet packets transform. According to the features between different channels of EEG signal, a multilayer one-dimensional convolutional neural network model was adopted to get a multi-class classification model for MI. This model was validated on the public datasets BCI competition IV 2a. The result shows that the mean accuracy of this model is 93.05%. This study is beneficial to MI-BCI research applications in the future.

Keywords- motor imagery; convolutional neural network; brain-computer interface; wavelet packet transform.

I. INTRODUCTION

Brain-computer interface (BCI) can establish a direct relationship between the human brain and machines, which is a valuable research topic for its potential to help disabled people to operate external devices directly [1]. This technology also has been concerned with rehabilitation therapy [2], automatic control [3], and entertainment[4]. An

essential procedure of BCI is to identify different mental tasks. Recently, motor imagery (MI) identification from electroencephalogram (EEG) signals based on BCI can accomplish this communication only with imagination, which is considered a promising technology for real application in the near future [5]. However, it is a tough problem to extract MI-EEG features for multi-class motor imagery classification because the EEG signals are non-stationary, non-linear, time-varying, and complicated, containing a great amount of time-frequency information. Many different approaches have been explored to solve this problem. Kumar et al. adopted the common spatial patterns (CSP) approach to acquire variance-based characteristics and used a deep neural network model for MI classification [6], but the CSP approach is susceptible to noise or interference and easy to lose time-frequency features. According to the convergence speed and classification accuracy, Zhang et al. selected the appropriate activation function and input the original EEG time-frequency image into the convolutional neural network (CNN) model with the short-time Fourier transform (STFT) method for classification [7], but it is hard to acquire both temporal and frequency resolution with this method. Xu et al. extracted the time-frequency features with adopted wavelet transform and applied these features to a 2-Layer CNN model obtaining a 90% accuracy result on the public datasets [8], but only low-frequency features were decomposed using this method, causing loss of information of some high-frequency features.

Deep learning neural network models are important tools for EEG signal feature extraction and have been widely adopted in EEG recognition. Dutta, K.K et al. adopted the

long short-term memory (LSTM) method to analyze the temporal characteristics of EEG signals and explored four-class MI classification tasks, but LSTM cannot perform parallel computation and the network computation is relative slow [9]. Sakhavi S. et al. used filter-bank common spatial patterns (FBCSP) to obtain the base feature matrix and used channel-wise convolution with channel mixing (C2CM) model [10] for secondary feature extraction and classification. With this method, a high classification accuracy of 74.46% was eventually obtained but the model parameters were computationally intensive and needed to be adjusted according to different subjects. Liu et al. extracted Spatio-temporal features of EEG signals based on a densely connected multi-branch 3D convolutional model [11] considering the characteristics of each dimension of EEG signal, but the model is computationally intensive.

In this study, a multi-class MI classification method for EEG signals based on wavelet packet transform and a one-dimensional convolutional neural network is presented. First of all, to reduce noise interference from the EEG signal acquisition procedure, data normalization and wavelet threshold denoise processes were applied to the original EEG motor imagery signals. Then, the wavelet transform method combining the time and frequency domains was used to process non-stationary time-varying EEG signals. To avoid significant loss of specific information and edge information with the traditional wavelet transform method which only can decompose the low-frequency part of EEG signals, the wavelet packet transform (WPT) method was applied to decompose both the low-frequency and high-frequency parts of EEG signals [12]. Moreover, wavelet packet nodes with μ (8-12Hz) and β (14-30Hz) rhythms were selected for EEG signal reconstruction. 1D-CNN model was used to extract and classify the features from reconstructed EEG signals. This model can make full use of the temporal features of EEG signals and convolve in time. With this method, temporal features of EEG signals can be fully extracted and be applicable to EEG signals with multiple input lengths. The parameters of the model are computationally cost-effective. Finally, the validation was performed with public datasets of BCI competition IV 2a. It was shown that the average multi-classification recognition rate of the method reached 93.05%, and the average Kappa value reached 0.8393. These validation results proved this method to be applicable for MI-BCI applications in the future.

II. MATERIAL AND METHODOLOGY

A. Dataset

In this study, the BCI Competition IV 2a data set was adopted to obtain the experiment data [13]. The International 10/20 system with 25 electrodes for 22 EEG channels and 3 EOG channels was used to get this dataset, and the EEG signals were recorded at 250 Hz sampling frequency, which were filtered with a band-pass filter from 0.5 to 100 Hz and trap filter at 50 Hz. Nine subjects were randomly assigned to imagine four styles of movements: left hand, right hand, feet, and tongue (denoted with 0, 1, 2, and 3). Each subject performed 2 rounds of experiments with 6 groups in each

round. Each group contained 48 times of motor imagery experiments. For every single motor imagery experiment, the experiment process is shown in Fig.1. The experiment started with a fixed cross on the screen and a brief prompt sound. After two seconds, an arrow of left, right, up, or down (denoted with a left hand, right hand, tongue, or feet) appeared on the screen for 1.25 seconds, prompting subjects to execute a motor imagery assignment. Six seconds later,

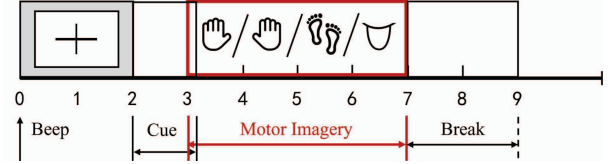


Figure 1. Process of motor imagery experiment.

the screen was shut down for a short rest period.

B. Normalization

To reduce the magnitude impacts of original data, these data were normalized to the same order of magnitude. With this method, the accuracy of the model can be increased because it can reduce the deviation between different feature values and facilitate accurate feature extraction. We applied the Z-Score normalization method and the formula was as follows:

$$x^* = \frac{x - \mu}{\sigma} \quad (1)$$

Where x is the original data, μ is the average value of original data, and σ is the standard deviation of original data, x^* is normalized data.

C. Denoising

Since the original EEG data usually contain noise from different kinds of interference, including industrial frequency interference, electromyographic interference, ECG artifacts, and so on. In this research, the wavelet threshold denoising method based on soft thresholding was adopted to remove these above-mentioned noises and the process is presented as follow:

1) *Wavelet Decomposition of EEG Signals*: Choose daubechies 4 (db4) waveform and the decomposition layer W , wavelet decompose the noise-containing signal to the W layer, and the equation of calculating the wavelet coefficient with the layer number w and the point number k was as follows:

$$WT_4^{(w,k)}(a, \tau) = \frac{1}{\sqrt{a}} \sum_{n=1}^N x_n \psi_4^{(w,k)}\left(\frac{n-\tau}{a}\right) \quad (2)$$

Where N is the number of EEG signal amplitudes, x_n is the amplitude of the n th EEG signal, $\psi_4^{(w,k)}((n-\tau)/a)$ is the db4 wavelet basis function of the node with layer number w

and the point number k , τ is the translation number, and $WT_4^{(w,k)}(a, \tau)$ is the wavelet coefficient of node (w, k) .

2) *Thresholding Wavelet Coefficients*: With a fixed threshold, the optimized threshold thr was calculated under the restriction of maximum and minimum estimation as follows:

$$thr = \sigma \sqrt{2 \ln N} \quad (3)$$

where σ is the noise standard variance, N is the signal length. The soft thresholding approach with the following equation was used to process decomposed wavelet coefficients.

$$s = \begin{cases} \text{sgn}(c)(|c| - thr) & |c| > thr \\ 0 & |c| \leq thr \end{cases} \quad (4)$$

where s is the signal after threshold processing, thr is a threshold value.

3) *Wavelet Inverse Transform*: The wavelet coefficients after the thresholding process were reconstructed to obtain the denoised EEG motor imagery signals.

D. Wavelet PacketS Transform

Wavelet packet transform can decompose both the low-frequency and high-frequency parts of EEG signals, avoiding the loss of features and EEG edging information, which can increase the accuracy of EEG motor imagery signals with time-frequency analysis. Wavelet packet nodes of μ (8-12Hz) and β (14-30Hz) rhythms were adopted in this work.

1) Wavelet Packet Transformation Decomposition

We set the wavelet packet tree node as (w, n) , w is the number of the layers, n is the wavelet packet node number and for k point number, the wavelet packet decomposition coefficients were calculated as follows:

$$d_w^{2n}(k) = \sum_{l \in Z} g_0(h-2k) d_{w-1}^n(l) \quad (5)$$

$$d_w^{2n+1}(k) = \sum_{l \in Z} g_1(h-2k) d_{w-1}^n(l) \quad (6)$$

Where, $g_0(h-2k)$ and $g_1(h-2k)$ are two conjugate orthogonal filters, and l is the number of filter coefficients. Wavelet packets decompose the original EEG signal into different nodes, with each node corresponding to a frequency range. If the signal sampling rate is f_s , the frequency bands of each node in the layer w were calculated as follows:

$$\left[0, \frac{f_s}{2^{w+1}}\right], \left[\frac{f_s}{2^{w+1}}, \frac{2f_s}{2^{w+1}}\right], \dots, \left[\frac{(2w-1)f_s}{2^{w+1}}, \frac{f_s}{2}\right] \quad (7)$$

2) Wavelet Packet Transformation Reconstruction

All nodes containing μ and β rhythms were selected for reconstruction. For a node with layer number $w+1$ and the point number k , then the wavelet packet coefficient was reconstructed as follows:

$$d_{w+1}^n(k) = \sum_{l \in Z} g_0(h-2k) d_w^{2n}(l) + \sum_{l \in Z} g_1(h-2k) d_w^{2n+1}(l) \quad (8)$$

E. Convolutional Neural Network Model

For multi-channel EEG signals, a suitable convolutional neural network model needs to be selected for training and learning according to the features of signals. Currently, CNN models have been extensively adopted and made a great performance in computer vision [14]. Meanwhile, CNN is capable of studying the input data expression, which can be adopted to study the properties of time-frequency in EEG motor imagery signals. We applied the 1D-CNN model to extract the internal features of each channel of the EEG motor imagery signals as follows.

1) Input Layer

EEG data of specific frequency bands extracted from the wavelet packet transform process were used as the input layer.

2) Convolution Layer

Since the one-dimensional convolution operation can acquire the significant local characteristics between the feature vector elements [15], we chose a one-dimensional convolution kernel. Moreover, the convolution kernel size was $1 \times q$, the weight of the row was 1, and the column j in the convolution kernel was w_{1j} . The amplitude of each element in the time-frequency matrix block corresponding to the convolution kernel was v_{1j} . The following equation was used to extract features of the next layer.

$$\text{conv}_{a,b} = \sum_j^q w_{1j} v_{1j} \quad (9)$$

where $\text{conv}_{a,b}$ was the value of the matrix node with row a and column b after the convolution operation. In this study, the kernel size of each convolution layer was set to 1×8 , 32 filters were used, the convolution step was 1 and the padding parameter was set as 0. During the convolution process, zero was added to the edge of the input matrix to prevent data loss. After that, the ReLU activation function was applied for network training with the formula as follows.

$$z_{a,b} = \max(0, \text{conv}_{a,b}) \quad (10)$$

Where $z_{a,b}$ is the activated matrix node value of the row a , the column b .

3) Pooling Layer

The pooling layer can decrease the range of feature matrix and parameter computation times. We chose max-pooling to emphasize the signal features.

4) Fully Connected Layer

The fully connected layer can help to reduce network size and improve nonlinear mapping capability [16]. In this study two completely connected layers were employed. The first layer had 400 neurons and the second layer had 4 neurons. As follow, the output vector U_l of the l layer in the fully connected layer was calculated.

$$U_l = U_{l-1}W_{l-1} + b_{l-1} \quad (11)$$

Where U_{l-1} and b_{l-1} are the weight matrix and bias of the layer. The final motion imagery output O was calculated as follows.

$$O = U_r W_r + b_r \quad (12)$$

Where r is the total number of layers, U_r , W_r and b_r are the output values of the last implied layer, the weight matrix, and the bias.

5) Output Layer

The probability distributions for each class were calculated with the Softmax function to predict the action of motor imagery EEG signals.

$$P_i = e^{x_i} / \sum_{j=1}^n e^{x_j} \quad (13)$$

Where P_i is the possibility of outputting the class i , $i = 1, \dots, n$, and n is the number of classification categories. x_i is the amount of i class output.

To decrease impacts from parameters setting, the parameters of the 1D-CNN model were finally determined through several experiments. In all, a 1D-CNN model was used in this study, and the diagram was shown in Fig.2, which has 3 convolution layers, 3 max-pooling layers, and 2 fully connected layers.

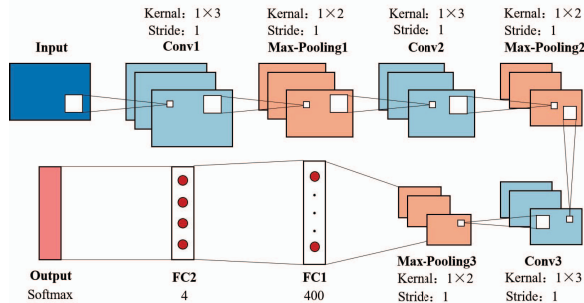


Figure 2. 1D-CNN structure for EEG motor imagery classification.

III. RESULTS AND DISCUSSION

A. Classification Results

Accuracy (Acc) and Kappa value were used to evaluate the performance of the trained model, which are commonly used in MI classification research. Accuracy is the ratio between the number of correctly classified samples and the total number of samples. Kappa value is calculated as follows, which can eliminate the effect of random classification [17].

$$Kappa = \frac{p_0 - p_e}{1 - p_e} \quad (14)$$

Where p_0 is the overall classification accuracy and p_e is calculated as follow.

$$p_e = \frac{r_1 \times c_1 + r_2 \times c_2 + \dots + r_m \times c_m}{n \times n} \quad (15)$$

Where n is the total number of samples. r_m and c_m denote the number of real and predicted samples in class m , respectively. The multi-class classification tests were carried out with BCI IV 2a public datasets. The results of the multi-class accuracy values of selected 9 subjects are shown and compared with other classical algorithms in Table I. Additionally, the kappa values are shown in Table II.

TABLE I. CLASSIFICATION ACCURACY OF DIFFERENT MODELS (%).

Subjects	CNN-LSTM [18]	FBCSP [19]	3D-CNN [20]	RBM-SVM [21]	This work
1	89.00	76.00	77.39	86.60	95.27
2	69.00	56.50	60.14	61.26	86.14
3	92.00	81.25	82.92	87.27	94.80
4	82.00	61.00	72.28	75.20	89.55
5	84.00	55.00	75.83	64.54	92.40
6	68.00	45.25	68.98	65.90	89.30
7	95.00	82.75	76.03	83.78	97.75
8	90.00	81.25	76.85	89.90	95.60
9	92.00	70.75	84.66	92.07	96.71
Mean	84.56	67.75	75.01	78.50	93.05

From Table I, the personal status of subjects will influence the quality of EEG motor imagery signals in the procedure of EEG motor imagery signals acquisition, which in turn affects the effect of classification recognition. For the same data, the multi-class accuracy of the proposed approach is higher than other classical approaches on the dataset of a single subject and the dataset of all subjects. The average accuracy was up to 93.05%, which is higher than other methods reported before proving the efficiency of this approach.

TABLE II. KAPPA VALUE OF DIFFERENT MODELS.

Subjects	CNN-LSTM [18]	FBCSP [19]	3D-CNN [20]	RBM-SVM [21]	This work
1	0.8500	0.6760	0.6990	0.8214	0.8478
2	0.5400	0.4170	0.4590	0.4838	0.7032
3	0.8700	0.7450	0.7880	0.7696	0.9024
4	0.7800	0.4810	0.5940	0.6664	0.8460
5	0.7700	0.3980	0.6470	0.5024	0.8298
6	0.6600	0.2730	0.5380	0.5301	0.7659
7	0.9500	0.7730	0.6530	0.7837	0.9255
8	0.8300	0.7550	0.7020	0.8655	0.8610
9	0.900	0.6060	0.7130	0.8942	0.8720
Mean	0.7944	0.5693	0.6437	0.7019	0.8393

Shown in Table II, the mean Kappa value of the proposed method was up to 0.8393 with high consistency for different subjects and was higher than other classical methods reported before.

B. Model Training discussion

To conduct the training experiments, experiment data from all subjects were combined together and was classified by a 5-fold cross-validation method. With this method, the original dataset was broken up into the training dataset and the test dataset for training and verification of classifiers. During the training process, the stochastic gradient descent (SGD) optimizer algorithm was applied to optimize the loss function. The training parameters used for the model are shown in Table III.

TABLE III. MAIN PARAMETERS OF THE CNN MODEL.

Parameter	Data
Optimizer	SGD
Epoch	100
Learning rate	0.01
Dropout	0.5
Batch size	24

C. Effect of Wavelet Packet Transform on Classification Results

Based on the same original data and other processing procedures, the effect of wavelet packet transformation on classification results was verified. The EEG data without wavelet packet transform was directly used as input parameters in the first group of experiments, and the EEG data after wavelet packet decomposition and reconstruction was used as input parameters in the second group of experiments. The EEG data of both groups were processed with the same data normalization, wavelet threshold denoising, and one-dimensional convolutional neural network procedures. Confusion matrix was used as the evaluation index in this experiment, and the classification results from two sets of experiments are shown in Fig.3 and

Fig.4. The non-diagonal values indicate the probability of incorrect classification for each category.

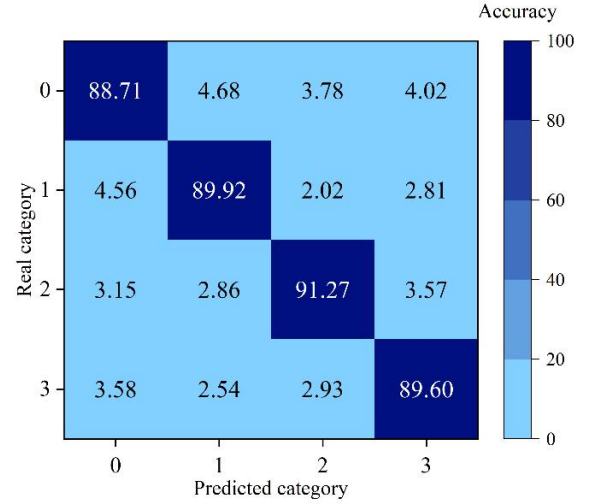


Figure 3. Classification accuracy of MI EEG without WPT.

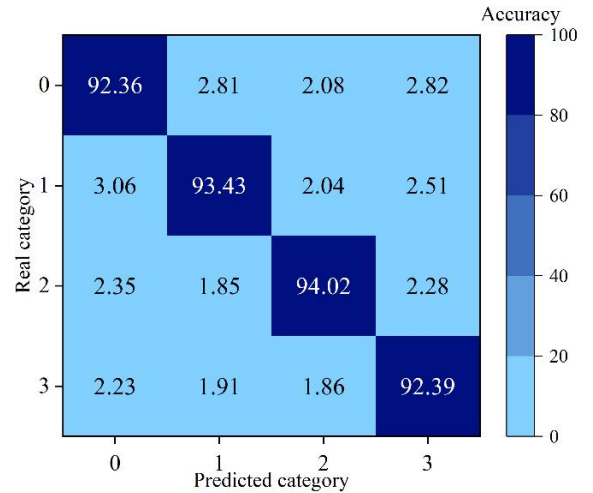


Figure 4. Classification accuracy of MI EEG processed by WPT.

From Fig.3 and Fig.4, the classification accuracy is higher with the wavelet packet transform method. This is because the wavelet packet transform can screen out the useful EEG motor imagery signals while avoiding the impact of invalid clutter information on classification.

IV. CONCLUSION

A multi-class classification method for MI EEG signals using wavelet packet transform and 1D-CNN was proposed in this study. Wavelet packet decomposition and reconstruction methods were applied for time-frequency analysis to extract MI information. With a trained 1D-CNN model, using one-dimensional convolutional kernels can acquire the internal information of different channels, which is helpful to enhance the multi-class MI classification accuracy up to 93.05%. The efficiency of this method has

been proved with public data sets. This method has the potential to be used in different kinds of MI-BCI applications.

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